

Victoria University of Wellington

Does a genetic reduction in the serotonin transporter affect the behavioural effects of ethanol?

Lisette Klein 2066513

Supervisor: Prof. B. Ellenbroek & dr. W.F.A. den Dunnen
July 2015
Major report BCN research master

Contents

Abstract.....	2
Introduction	3
Material and methods	6
Results.....	8
Discussion.....	11
References	16
Appendix	20
<i>Material & methods</i>	20
<i>Results</i>	20
<i>Discussion</i>	20
Protocol ethanol self-administration	21
Protocol locomotion behaviour	21
Protocol rotarod.....	21
Protocol blood plasma preparation	23
Protocol Ethanol Essay Kit.....	23

Abstract

Ethanol is the most widely consumed drug of abuse used by the human population and is an important cause of health issues associated with a high socioeconomic impact. In the brain, alcoholism is associated, among others, with a decreased activity of the serotonin transporter (SERT), which facilitates re-uptake of centrally released serotonin (5-HT). Given that alcoholism in humans has a genetic component, we investigated whether rats with a genetic reduction in SERT show a differential sensitivity to the behavioural properties of ethanol. For that reason we studied ethanol self-administration, as well as ethanol induced ataxia and changes in locomotor activity in male and female serotonin transporter knockout rats.

There was an interaction between genotype and ethanol dose in the level of ethanol-induced ataxia. SERT knockout rats did better relative to both heterozygous and wild-type rats and were less influenced by the ataxic effects of ethanol. Especially, a dose of 2.0 g/kg ethanol had a detrimental effect on motor coordination in heterozygous and wild-type rats, but not in SERT knockout animals.

In a novel environment, SERT knockout animals showed a higher number of ambulatory counts compared to their counterparts. 1.0 g/kg ethanol induced a decrease in activity for all genotypes compared to saline or 0.25 g/kg.

Ethanol self-administration was similar in SERT knockout and wild-type animals. When animals had 7 h access to ethanol, females showed a higher intake and preference for ethanol compared to males. However, when the period was extended to 24 h, males showed a greater increase in ethanol intake compared to the first 7 h period.

The data suggest that animals homozygous for the SERT gene are less likely to show ethanol-induced ataxia. However, they do not show a differential sensitivity to the rewarding properties of ethanol or ethanol-induced locomotion. Therefore, the results are likely to reflect recreational ethanol intake, rather than dependence.

Key words: Serotonin transporter knock-out rat model; Intermittent access schedule; Ethanol-induced locomotor activity; Ethanol-induced ataxia; Rotarod

Introduction

Addiction is characterized by an uncontrollable desire for rewarding stimuli, which is maintained despite distress or a negative impact on the subject. An addiction is composed out of three different phases with specific neurobiological adaptations. The phases are: intoxication, withdrawal and anticipation (George F. Koob & Volkow, 2010). The first phases of addiction development are mainly driven by positive reinforcement. However, positive reinforcement decreases over time due to reductions in neurotransmitter function that are associated with reward. In the long run, drug-abuse is maintained by negative reinforcement, which is the increased likelihood of a response when an aversive state is removed.

A rewarding substance that can lead to addiction is ethanol. Ethanol has both a stimulating and a sedating influence on an organism. Low doses are stimulating, whereas high doses sedate the central nervous system. In the brain, ethanol can be found within minutes after intake and it exerts an effect on the neurotransmission between neurons. One of the acute effects of ethanol in the brain is an increase of the extracellular 5-HT levels in the nucleus accumbens (NAc), which plays an important role in reward processing (Weiss et al., 1996). However, chronic ethanol intake and withdrawal lead to decreased 5-HT levels in the NAc (Kirby et al., 2011; Weiss et al., 1996). In rhesus monkeys the binding of [18F]mefway to the 5-HT_{1A} receptor was amplified during ethanol drinking (Hillmer et al., 2014), indicating an increased expression of the receptor in response to ethanol. Ethanol reduces the activity of the NMDA receptor, which is responsive to excitatory glutamate, and decreases memory formation by inhibiting long term potentiation in the hippocampus. In contrast, ethanol increases the activity of the inhibitory GABA_A receptor and the serotonin 5-HT₃ receptor. However, chronic ethanol exposure increases activity of NMDA receptor and voltage-gated calcium channels, which can lead to excitotoxicity

(Diamond & Gordon, 1997). The GABA_A receptor function decreases after chronic ethanol exposure. These changes in neurotransmitter function are adaptive and long-lasting.

Another neurotransmitter that plays a role addiction is serotonin. Serotonin, or 5-hydroxytyptamine (5-HT), is a neuromodulating transmitter that plays an essential role in reward-related processes, ranging from feeding to sexual behaviour (Hayes & Greenshaw, 2011). 5-HT is produced by the raphe nuclei, which can be found in the brain stem. The raphe neurons project to nearly the entire brain and have a modulating function on all major neurotransmitters, including the other reward-related neurotransmitter dopamine (Hayes & Greenshaw, 2011; Kirby, Zeeb, & Winstanley, 2011; Lovinger, 1997). 5-HT release is associated with, and altered by, the value of expected and received rewards (Kranz, Kasper, & Lanzenberger, 2010).

Once 5-HT is released into the synaptic cleft, reuptake will take place by the presynaptic neuron through the serotonin transporter (5-HTT or SERT). This makes the 5-HTT a regulator of the duration of the synaptic action of 5-HT (Gobbi, Murphy, Lesch, & Blier, 2001; Lovinger, 1997). A 5-HTT-linked polymorphic region (5-HTTLPR) in the promoter region of the 5-HTT is present in the human population. More than 40% of the European population has a shorter version of the promoter, the so-called S-allele. This S-allele has a 44 base pair deletion and leads to a decreased activity of 5-HTT gene transcription (Heils et al., 1996; Lesch et al., 1996). Moreover, 5-HTT mRNA levels (Little et al., 1998), availability of transporters in raphe nuclei (Heinz et al., 1998), and re-uptake ability (Lesch et al., 1996) are reduced, resulting in increased extracellular 5-HT levels.

The 5-HTT S-allele is associated with a higher vulnerability for development of a substance use disorder. Cao et al. (2013) performed a meta-analysis on 38 studies

investigating ethanol abuse. The authors found a significant association between the 5-HTT S-allele and severe alcoholism in a combined European and Asian population (Cao, Hudziak, & Li, 2013). Moreover, the S-allele is associated with ethanol dependence (Feinn, Nellisery, & Kranzler, 2005) and a higher initial level of intake (van der Zwaluw et al., 2010). 19% of the New-Zealand population have a hazardous drinking pattern (Ministry of Health, 2013). A hazardous drinking pattern is characterized by a high frequency, quantity and irregular timing of drinking (Babor, Higgins-Biddle, Saunders, & Monteiro, 2001). Ethanol abuse is associated with a decreased 5-HT turnover, caused by the decreased re-uptake of 5-HT (A. Heinz et al., 2003).

Both underactivity and overactivity of 5-HT can cause problems, such as anxiety and depression. The adverse effects of 5-HT activity can therefore be visualised as an inverted U-shaped function (Carnicella, Yowell, & Ron, 2011; Olivier et al., 2008). This inverted U-shape might also apply for ethanol drinking in relation to 5-HT transmission. In the European, Asian, and Mexican population the S-allele resulting in high levels of extracellular 5-HT was the risk factor for drug abuse, while for the African population it was the L-allele, which results in higher reuptake of 5-HT (Cao et al., 2013).

The SERT knock-out rats (SERT^{-/-}) were generated by N-ethyl-N-nitrosourea (ENU)-driven target-selected mutagenesis (Smits et al., 2006). The mutation is a pre-mature stop-codon in the exon coding for the second extracellular loop of the transporter. The result is a nonsense mutation, which leads to the absence of an mRNA transcript and protein in the SERT^{-/-} and a 50% reduced expression of both in SERT^{+/-}. As a consequence of the nonsense mutation, a 9-fold increase in the extracellular levels of 5-HT is found in the SERT^{-/-} compared to their wild-type (SERT^{+/+}) littermates. Moreover, intracellular levels of 5-HT and depolarization-induced 5-HT release were reduced in the

SERT^{-/-} rats. However, the 5-HT uptake was only completely eliminated in the SERT^{-/-} after blocking of the noradrenaline transporter, indicating that there is some redundancy at the level of reuptake transporters (J. Homberg et al., 2007). The high extracellular levels of 5-HT in the SERT^{-/-} rats can initiate desensitization of the 5-HT receptors on the post-synaptic neurons, which would lead to decreased 5-HT neuronal transmission. Homberg et al. (2008) found that SERT^{-/-} rats showed decreased binding of [³H]WAY-100635, a 5-HT_{1A} antagonist, in the dorsal raphe nuclei, CA2 and CA3 regions of the hippocampus, and in the cingulate cortex. Importantly, WAY-100635 is not completely selective for the 5-HT_{1A} receptor; it can also bind the D4 receptors of the dopaminergic system (Chemel, Roth, Armbruster, Watts, & Nichols, 2006). Therefore, it could be that the 5-HT receptors are even more down regulated in the SERT^{-/-}, which could not be shown due to the non-selectivity of the marker. Despite a desensitisation of the inhibitory 5-HT_{1A} auto receptors, mice lacking the SERT displayed a decreased neuronal firing activity of dorsal raphe nuclei (Gobbi et al., 2001).

The current study will use an intermittent ethanol access schedule. This involves repeated episodes of free-choice ethanol intake and withdrawal and is therefore a model of ethanol abuse (Carnicella, Ron, & Barak, 2014). This model has several advantages over other methods. First of all, the rats will voluntarily consume ethanol without any initiation events (Simms et al., 2008; Wise, 1973). Moreover, the animals exhibit alterations in their behaviour, and on a molecular and cellular level, which is different from the effects induced by passive administration (Jacobs, Smit, de Vries, & Schoffemeer, 2003). The intermittent ethanol access is suggested to induce changes in the mesolimbic system; in particular dopamine levels are reduced after a withdrawal period of 24 h. This decreased level of dopamine is linked to the craving for ethanol before relapse (Carnicella et al., 2014). Taken

together, this protocol leads to a good representation of the human situation.

Innate differences in the neurotransmitter systems may contribute to an individual's vulnerability to the reinforcing properties of ethanol. The different neurotransmitter receptors have different expression patterns throughout the brain, which results in a differential effect of ethanol for some brain regions (Vengeliene et al., 2008). Phosphorylation of receptors plays a likely regulating role in ethanol sensitivity (Diamond & Gordon, 1997). Serotonin amplifies ethanol-induced excitation in the ventral tegmental area resulting in dopamine release and the experience of reward (Brodie, Trifunović, & Shefner, 1995). Besides, ethanol has an effect on opioid peptides, which can enhance the rewarding properties of it (George F Koob & Le Moal, 2005). Chronic ethanol intake reduces cAMP production by desensitization of receptors responsible for the synthesis. Moreover, deficits in cAMP responsive element binding protein (CREB), an important mediator of neuronal changes in transcription, in the central amygdala are associated with ethanol dependence (Moonat, Starkman, Sakharkar, & Pandey, 2010). At very high concentrations ethanol might even change the membrane composition (Vengeliene et al., 2008) and thereby impair cell functioning.

The SERT^{-/-} rats have high levels of 5-HT and can therefore be very sensitive to the rewarding properties of ethanol. On the other hand, the high levels of 5-HT in the SERT^{-/-} can make them less sensitive to ethanol, due to compensatory down regulation of 5-HT_{1A} and 5-HT_{2B} auto receptors in the raphe nuclei and substantia nigra found in SERT^{-/-} mice (Fabre et al., 2000) and rats (Homberg et al., 2008). These compensations are likely to be a result of a homeostatic mechanism to normalize changes (Meredith, 2015). Compensation is likely to occur in human subjects as well, but it is unclear whether they occur to similar extent and in the same systems.

The SERT^{-/-} animals show an increased

sensitivity to the reinforcing properties of other drugs of abuse, such as MDMA (Oakly, Brox, Schenk, & Ellenbroek, 2014) and cocaine (Homberg et al., 2008). However, the animals do not show an increased sensitivity for heroin (Brox & Ellenbroek, unpublished results). The different classes of drugs have different mechanisms in the brain. MDMA and cocaine are psychostimulants and release dopamine in an impulse-independent way, meaning that they block and/or reverse dopamine reuptake rather than increasing dopamine firing at the level of the cell body (Kuhar, Ritz, & Boja, 1991; Prus, James, & Rosecrans, 2009). On the other hand, heroin activates dopamine firing and dopamine release is therefore impulse-dependent (Prus et al., 2009). Moreover, heroin and ethanol exert their reinforcing properties by the endogenous cannabinoid system, while cocaine does not (Caillé, Alvarez-Jaimes, Polis, Stouffer, & Parsons, 2007).

The aim of this study is, therefore, to investigate whether rats with a genetic reduction in SERT show a differential sensitivity to the rewarding properties of ethanol administration. For that reason ethanol self-administration by the intermittent access schedule will be used, because it is a model for human addiction. Besides, ethanol-induced ataxia or motor incoordination will be assessed by the accelerating rotarod and changes in locomotor activity will be evaluated in a novel environment, because these measures are easy to quantify and translate to the human situation. Both male and female serotonin transporter knockout rats will be used to investigate possible sex-differences, because 1 in 5 men develops ethanol addiction whereas this is 1 in 11 for women. Therefore, it is expected that males will consume more ethanol compared to females. Based on the association of a reduction in SERT and ethanol dependence in humans, it is hypothesized that the SERT^{-/-} will show a differential sensitivity for the rewarding properties of ethanol.

Material and methods

Experiment 1 - Ethanol consumption

Animals

33 adult, ethanol-naïve, male and female SERT^{+/+} and SERT^{-/-} (Slc6a4^{1Habr}) Wistar rats (Smits et al., 2006). 15 (8 SERT^{+/+} and 7 SERT^{-/-}) males weighting 320-500g and 18 (9 SERT^{+/+} and 9 SERT^{-/-}) females weighting 200-300 g were used for the first part of this study. All subjects were bred at the Victoria University of Wellington, New Zealand. Animals were housed 2 or 3 per cage in standard cages in a temperature-controlled (21±1°C) and humidity-controlled (60%) room and had food (Diet 86, Sharpes stockfeed, Carterton, NZ) and water ad libitum. The room was subject to a 12-hour reversed light/dark cycle with lights off at 7.00 AM. The animals were allowed to acclimatize to the reversed light/dark cycle and handling for 2 weeks. Afterwards they were housed individually to be able to measure fluid intake. All experimental procedures were approved by the Animal Ethics Committee of Victoria University of Wellington (application 21549).

Ethanol intake procedures

Fluids were offered in 120 ml graduated plastic bottles with stainless steel, drip proof, double ball bearing drinking sprouts (Living World #61530, Baie d'Urfé, Quebec, Canada) inserted at the front of the cage 30 minutes after lights off in the reversed light/dark cycle room. Total fluid intake was measured by weighting the bottles prior to and after each session of 7 or 24 hours, also on the water

days. Animals were weighted daily Monday through Friday to be able to calculate ethanol intake in grams per kilogram of body weight and total fluid intake per gram of body weight. The preference for ethanol over water was calculated as the ratio of ethanol to total fluid intake. The 20% ethanol solution was acquired by diluting 99.8% (v/v) ethanol, with the highest degree of purity and without additives (PureScience Ltd, Porirua, New Zealand), in tap water. An empty cage with two bottles containing water and 20% ethanol were used to measure the leakage, which was on average 9.0% of total fluid intake (Carnicella et al., 2011).

Intermittent ethanol access schedule

The intermittent access to 20% ethanol drinking paradigm was adapted from Simms et al. (2008). Animals had access to ethanol during three 7-hour-sessions per week for four weeks (12 sessions), followed by three 24-hour-sessions per week for the next four weeks (12 sessions). On the Monday after the acclimatization period, each ethanol-naïve rat was given access to 1 bottle of 20% (v/v) ethanol and 1 bottle of water. After 7 hours, the ethanol bottle was weighted and replaced by a second water bottle in the 7-hour-session or placed back in the 24-hour-session. After 24 hours, the ethanol bottles was weighted again and replaced by a second water bottle during the 24-hour-session. On Wednesdays and Fridays this pattern was repeated. The rats had access to 2 water bottles over the weekend. The placement of the ethanol bottle in each drinking session was alternated to

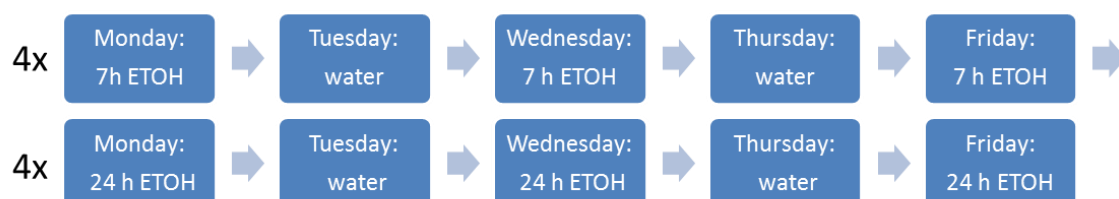


Figure 1. Timeline ethanol self-administration. The intermittent ethanol access schedule used in this study is outlined above. Animals had 7 h access to a 20% ethanol solution for 3 days a week (Monday, Wednesday and Friday), while the remaining time two water bottles were present. This schedule was maintained for 4 weeks. Afterwards the schedule remained the same, but the period ethanol was available was extended to 24 h per day on Monday, Wednesday and Friday. This schedule was maintained for another 4 weeks. 6

control for the development of side preferences.

Experiment 2 - Ethanol-induced locomotor activity

Locomotor activity was measured in an additional set of 105 ethanol-naïve male SERT^{+/+} (35), SERT^{+/-} (35) and SERT^{-/-} (35) rats (390-520 g). Animals were housed 2 or 3 per cage in standard cages in a temperature-controlled (21±1°C) and humidity-controlled (60%) room and had food and water ad libitum. The room was subject to a 12-hour light/dark cycle with lights on at 7.00 AM. Animals were introduced into the experimental room 10 min prior to an intraperitoneal (i.p.) injection with 0.9% saline, 0.25, 0.5, 1.0 or 2.0 g/kg ethanol and placed in a standard, transparent Plexiglas, open field (Med Associates, 43.2*43.2*30.5 cm) provided with infrared beams in sound-attenuated boxes for the period of 60 min. Activity was recorded between 9.00 AM and 12.00 PM in the dark and assessed by a computer program (ACTIVITY MONITOR Version 5 program; Med Associates Inc., St. Albans, VT, USA). Only the first 30 min (novelty-induced locomotor activity) were used for analysis, because after this the subjects were mainly inactive. Horizontal locomotor activity was analysed as ambulatory counts, and time spent in the central zone (CZ, the inner 19x19 cm of the box) was analysed as a measure for anxiety. A white noise generator covered extraneous auditory disturbance during testing, and the room was lightened by a red light (Colussi-Mas & Schenk, 2008). The apparatus was cleaned with dissolved Virkon S (Antec International, Sudbury, UK) in water (1:100). After more than 10%, corresponding to a number of 17 animals died following a dose of 2 g/kg, this dose was excluded for further use and the concentration of ethanol was adjusted to 50%.

Experiment 3 – Ethanol-induced ataxia

The male rats used in the ethanol-induced locomotor activity, described above, were also used for the investigation of ethanol-induced ataxia. A minimum of 8 weeks inter-experiment interval was set to assure a minimal carryover effect. 24 animals per genotype were trained to balance on a rotating roll (Rotarod LE 8500 (76-0239), 36*24*50 cm, Panlan, Barcelona, Spain). The rotating roll had a diameter of 60 mm and was equipped with 4 lanes each 75 mm in width. Each lane had a lever that registered the time and speed at which the animal fell from the roll. 10 min prior to testing, the animals were introduced into the room to adjust to the environment. The first day during habituation, the animals were placed on the roll without turning it on. The following three days each rat was given 3 trials a day with an intertrial interval of 5-10 min. During the training the apparatus was set at a fixed speed, which was build up from 4 to 12 rpm over the consecutive trials. The maximum time on the roll was 5 min. 48 h after the last training trial animals received an i.p. injection with 0.9% saline, 1.0 or 2.0 g/kg 50% (v/v) ethanol. 10 and 30 min following ethanol administration motor behaviour was recorded using the accelerating mode, increasing from 4 to 40 rpm over a time span of 5 min. Between each trial the apparatus was cleaned of faeces and after each session of 3 trials the apparatus was thoroughly cleaned using 70% ethanol. Latency to fall calculated as the average of the testing trials, and relative latency to fall (testing trials/last 3 training trials) were used as a measure for the degree of ataxia induced by ethanol.

Statistics

Statistical analysis was performed using SPSS version 22 (IBM, New York, USA). All collected data were checked for normality and homogeneity. The ethanol consumption was analysed using a mixed two-way ANOVA with repeated measures with gender and genotype as independent variables. The locomotor and ataxia measurements was analysed using a

two-way ANOVA with genotype and ethanol dose as independent variables. The Tukey post-hoc analysis was performed when a significant overall main effect was found with ANOVA ($p < 0.05$). The Huynh-Feldt test for within-subject effects was used whenever the sphericity assumption was violated. Pearson correlation analysis was performed on the variables interacting with each other. Data are expressed as mean $\pm$ SEM.

Results

Experiment 1- Ethanol consumption

The rewarding properties of ethanol were tested using the intermittent ethanol access schedule. Repeated measures ANOVA showed that there was no genotype effect or genotype $\times$ gender interaction in ethanol intake. However, a significant gender effect ($F_{(1, 29)} = 6.65$, $p = 0.015$) could be observed. As shown in Figure 2A overall consumption was

higher in female rats compared to males (2.96 ± 0.21 vs 2.15 ± 0.23 g/kg). Moreover, especially in the 7 h access period females consumed higher amounts of ethanol compared to males ($F_{(1, 31)} = 41.71$, $p < 0.001$; 2.71 ± 0.16 vs 1.25 ± 0.17 g/kg respectively).

After 12 sessions of 7 h access to the 20% ethanol solution, 12 sessions of 24 h followed. The transition from 7 h access to 24 h access resulted in an increased intake for the males ($F_{(6.48, 187.84)} = 16.03$, $p < 0.001$), but not for the females. Genotype ($F_{(6.48, 187.84)} = 2.41$, $p = 0.026$) also had an effects on the increased intake when the transition from 7 h access to 24 h access occurred, especially $SERT^{+/+}$ animals showed a greater increase in ethanol intake (1.95 ± 0.16 vs 3.48 ± 0.31) compared to $SERT^{-/-}$ animals (2.01 ± 0.17 vs 2.76 ± 0.32). No interaction could be observed between genotype and gender.

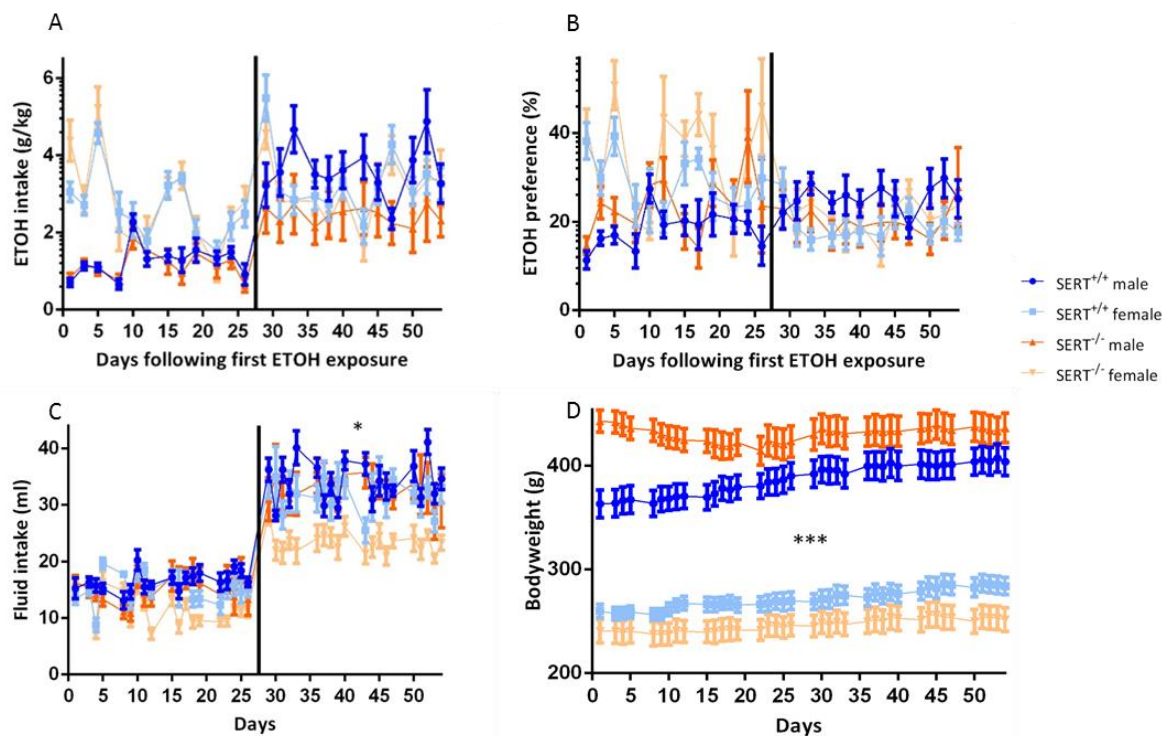


Figure 2. The intermittent access schedule results in **A.** higher ethanol intake in females compared to males (2.96 ± 0.21 vs 2.15 ± 0.23). **B.** the preference for ethanol over water did not differ between groups. **C.** Fluid intake was higher in males compared to females (25.28 ± 1.12 vs 20.75 ± 1.02), but **D.** bodyweight shows a significant difference between males and females (410.28 ± 8.29 vs 259.10 ± 7.55). The values are expressed as mean ethanol intake (g/kg), preference (ratio of ethanol over total fluid intake), fluid intake (ml), or bodyweight (g) $\pm$ SEM. * $p < 0.01$ fluid intake over 24 h compared to 7 h, ** $p < 0.001$ weight males compared to females. $n = 9$ for $SERT^{+/+}$ and $SERT^{-/-}$ females, $n = 8$ for $SERT^{+/+}$ males and $n = 7$ for $SERT^{-/-}$ males. (ETOH= ethanol; $SERT^{+/+}$ = wild-type; $SERT^{-/-}$ = homozygous).

The preference for the ethanol solution over water is depicted in figure 2B and did not differ between groups. Especially in the 24 h access period the preference stabilized around $21.58 \pm 1.58\%$ for all animals. $SERT^{-/-}$ animals decrease their preference for ethanol more when ethanol access was expanded compared to $SERT^{+/+}$ animals ($F_{(10.72, 310.94)} = 2.49$, $p = 0.006$). Moreover, females showed a bigger decrease in preference in the 24 h access sessions compared to males ($F_{(6.48, 187.84)} = 11.02$, $p < 0.001$).

Total fluid intake is illustrated in figure 2C and revealed a gender ($F_{(1, 29)} = 9.01$, $p = 0.005$) and genotype ($F_{(1, 29)} = 5.08$, $p = 0.032$) effect. Besides, a trend toward an interaction between gender and genotype ($F_{(1, 29)} = 3.40$, $p = 0.076$) could be observed. $SERT^{+/+}$ animals showed an overall higher consumption of fluids compared to $SERT^{-/-}$ animals. Moreover, males showed a higher intake of fluids compared to females. Total fluid intake on ethanol days compared to water days revealed an interaction between genotype and gender ($F_{(1, 29)} = 7.75$, $p = 0.009$). $SERT^{+/+}$ males and $SERT^{-/-}$ females consumed more fluids on ethanol days compared to water days. $SERT^{-/-}$ males and $SERT^{+/+}$ females consumed comparable amount of fluids on ethanol days compared to water days (not shown).

The relative fluid intake calculated by dividing total fluid intake by bodyweight showed a gender ($F_{(1, 28)} = 13.18$, $p = 0.001$) and a genotype effect ($F_{(1, 28)} = 4.38$, $p = 0.045$), but no interaction ($F_{(1, 28)} = 1.89$, $p = 0.180$). Females consumed relatively higher amounts of fluid than males. $SERT^{+/+}$ had a higher relative fluid intake compared to $SERT^{-/-}$ (not shown).

The expansion from 7 h to 24 h resulted in a genotype ($F_{(11.88, 344.63)} = 2.19$, $p = 0.012$), gender ($F_{(11.88, 344.63)} = 4.29$, $p < 0.001$) and an interaction between genotype and gender ($F_{(11.88, 344.63)} = 2.29$, $p = 0.009$) for the total fluid intake. $SERT^{+/+}$ showed a bigger increase in fluid intake after the transition

from 7 h access to 24 h access compared to $SERT^{-/-}$ animals. In addition, males increased their intake more over time in comparison to females. When both genotype and gender are taken into account, $SERT^{+/+}$ males show the biggest increase in fluid intake, followed by $SERT^{-/-}$ males, $SERT^{+/+}$ females, and $SERT^{-/-}$ females.

Bodyweight was measured daily which allowed the calculation of ethanol intake per kg and is shown in figure 2D. Bodyweight showed a significant gender effect ($F_{(1, 29)} = 181.98$, $p < 0.001$), but no genotype effect. As seen in the figure, males were heavier compared to females (410.28 ± 8.29 vs 259.10 ± 7.55 respectively). Moreover, an interaction between genotype and gender was observed for bodyweight ($F_{(1, 29)} = 10.02$, $p = 0.004$). $SERT^{-/-}$ males were the heaviest, followed by $SERT^{+/+}$ males, $SERT^{+/+}$ females, and $SERT^{-/-}$ females. Pearson's correlation revealed a significant positive correlation between bodyweight and fluid intake (0.490 , $p = 0.004$).

Experiment 2 – Ethanol-induced locomotor activity

The locomotor activity as a result of 0.25, 0.5, and 1.0 g/kg ethanol were assessed in a novel environment. Both genotype ($F_{(2, 72)} = 19.65$, $p < 0.001$), and dose ($F_{(3, 72)} = 5.21$, $p = 0.003$) had a significant effect on total number of ambulatory counts, but there was no interaction. As shown in figure 3A, $SERT^{-/-}$ animals have an overall higher activity (1180.93 ± 211.16) compared to $SERT^{+/+}$ (686.39 ± 90.41) and $SERT^{-/-}$ animals (547.39 ± 103.18) ($p < 0.001$). 1.0 g/kg ethanol (522.24 ± 110.57) resulted in a decreased activity compared to saline ($p = 0.003$; 964.05 ± 153.09) and 0.5 g/kg ($p = 0.011$; 915.14 ± 165.76), whereas 0.25 g/kg (818.19 ± 110.24) did not differ from any of the other doses.

Time spent in the central zone is indicative for anxiety. No differences between genotypes, dose or an interaction between genotype and dose were apparent.

Death rate and ethanol dose

Since several animals did not survive a dose of 2.0 g/kg 100% ethanol, this dose was excluded and the concentration was lowered to 50% ethanol. Investigation of the relationship between ethanol dose and death indicated a significant positive correlation (0.583, $p < 0.001$; not shown), which was also present without the 2.0 g/kg group (0.311, $p = 0.004$). This shows that a higher dose resulted in a higher death rate. Moreover, a negative correlation became apparent between death and concentration of injected ethanol (-0.219, $p = 0.036$), suggesting that a lower concentration is associated with a higher survival rate.

Experiment 3 – Ethanol-induced ataxia

Ataxia was measured as the latency to fall from an accelerating rotating roll. The testing phase showed a genotype ($F_{(2, 63)} = 8.19$, $p = 0.001$), a dose ($F_{(2, 63)} = 7.92$, $p = 0.001$), and an interaction between genotype and ethanol dose ($F_{(4, 63)} = 3.86$, $p = 0.007$). $SERT^{-/-}$ animals that received saline were able to remain the longest on the accelerating roll compared to $SERT^{+/+}$ and $SERT^{+/-}$ animals that received saline ($F_{(2, 21)} = 8.45$, $p = 0.002$; figure 4A).

Moreover, $SERT^{-/-}$ animals that either received 1.0 g/kg or 2.0 g/kg perform less well compared to $SERT^{-/-}$ animals that received saline ($F_{(2, 21)} = 7.73$, $p = 0.003$). Pearson correlation revealed a strong positive correlation between latency to fall and speed at which it occurred (0.983, $p < 0.001$). There is no correlation between latency to fall and bodyweight.

The relative latencies to fall were calculated by dividing the testing trials by the last 3 training trials, because the absolute data showed big individual differences. These data show a genotype ($F_{(2, 63)} = 5.83$, $p = 0.005$) and a dose effect ($F_{(2, 63)} = 6.02$, $p = 0.004$), but no interaction. Figure 4B shows the relative latencies to fall and shows that saline treated animals perform comparable to baseline (dashed line at 100%). $SERT^{+/-}$ animals that received 1.0 g/kg decrease in performance compared to baseline ($F_{(1, 7)} = 31.04$, $p = 0.001$), $SERT^{+/+}$ and $SERT^{-/-}$ did not. 2.0 g/kg resulted in a significant decrease in the latency to fall for both $SERT^{+/+}$ ($F_{(1, 7)} = 12.85$, $p = 0.009$) and $SERT^{+/-}$ ($F_{(1, 7)} = 55.17$, $p < 0.001$), but not for $SERT^{-/-}$ animals.

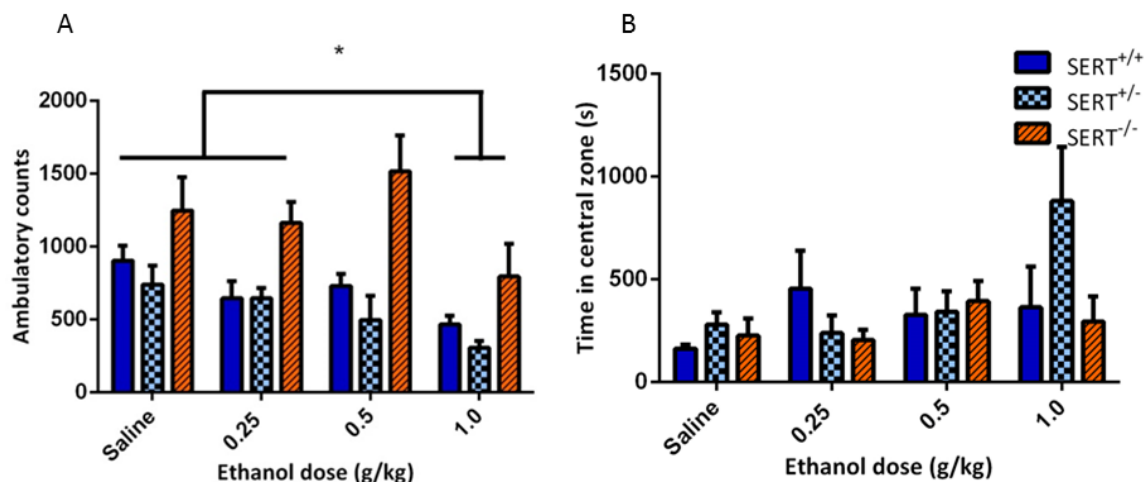


Figure 3. Ethanol-induced locomotor activity is **A**. decreased by a medium dose (1.0 g/kg) of ethanol (522.24 ± 86.74). Moreover, $SERT^{-/-}$ animals show an increased overall activity (1180.93 ± 75.12). **B**. time spent in the central zone was not different between the groups. Values are expressed as mean ambulatory counts, or time in central zone (s) $\pm$ SEM. * $p < 0.01$ 1.0 g/kg vs saline and 0.25 g/kg. ($SERT^{+/+}$ = wild-type; $SERT^{+/-}$ = heterozygous; $SERT^{-/-}$ = homozygous) $n = 7$ per group.

Discussion

This study investigated the differential sensitivity of the SERT knockout rat model to the behavioural properties of ethanol. The results showed that the SERT^{-/-} are less sensitive for ethanol-induced ataxia and showed a higher overall locomotor activity. On the other hand, ethanol self-administration was comparable between SERT^{-/-} and SERT^{+/+} animals.

Experiment - 1 Ethanol self-administration

Ethanol self-administration using the intermittent ethanol access schedule showed a similar ethanol intake in the SERT^{-/-} and SERT^{+/+} animals. This schedule resulted in a consumption of 3.48 g/kg/24 h in the SERT^{+/+} in contrast to the findings by Simms et al. who report a baseline consumption of 5.8 g/kg/24 h (Simms et al., 2008). However, Simms et al. start with 24 h access, contrary to the 4 weeks of 7 hour access we used prior to 24 hour sessions. However, a study by a collaborator used the same protocol we used and found an ethanol intake of 2 g/kg/24 h, which is even lower than our findings (Lesscher, personal communication). Another study using SERT^{-/-} mice showed a lower unconstrained demand

in ethanol self-administration (Lamb & Daws, 2013). However, this study used a different concentration of ethanol. Moreover, our data are comparable to a study in SERT^{-/-} mice where the authors also did not find a difference in ethanol intake and preference between SERT^{-/-} and SERT^{+/+} animals (Boyce-Rustay et al., 2006). However, that study used ethanol concentrations ranging from 3-11% and we used 20%. It is likely that lower concentrations are less reinforcing than higher concentrations (Meisch & Thompson, 1971).

Interesting is the low preference of only 21.58% for the ethanol solution over water found in this study and the decrease observed when session length was increased from 7 h to 24 h. Other studies that investigated ethanol consumption report a preference of at least 40% (Boyce-Rustay et al., 2006; Simms et al., 2008; Wise, 1973). Some of these studies used 20% ethanol and others used a lower concentration. The animals were housed in a shared room, which implied that other researchers entered the room on several occasions during the day. This might have a disturbing impact on the animals leading to a low intake of ethanol. Moreover, as shown in figure 1A & B,

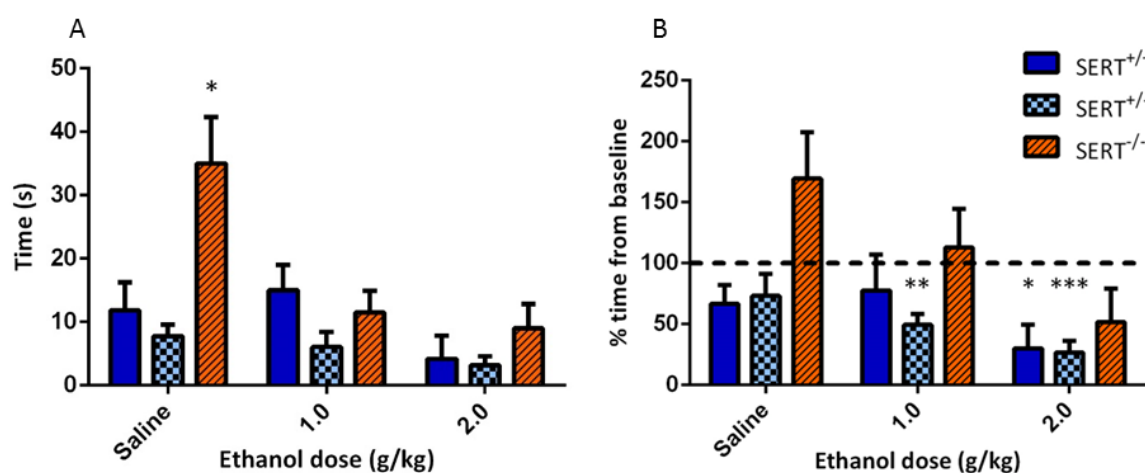


Figure 4. Ethanol-induced ataxia is **A.** most apparent in SERT^{-/-} animals, which show the biggest difference between performance under saline and 2.0 g/kg conditions **B.** the relative latency to fall is decreased for SERT^{+/-} receiving 1.0 and 2.0 g/kg and SERT^{+/+} receiving 2.0 g/kg ethanol. Values are expressed as mean time (s), or relative time (%) $\pm$ SEM; * p < 0.05, ** p < 0.01 and *** p < 0.001 compared to the other groups (A) or baseline performance (B). (SERT^{+/+} = wild-type; SERT^{+/-} = heterozygous; SERT^{-/-} = homozygous) n = 6 per group.

there was some variation in intake between days, especially in females. Unfortunately, males and females were tested in different cohorts due to shortage in resources. Variations in intake between days might be due to deviations in the quality of the ethanol solution. Moreover, variations in external disturbance might be different between days. It is most likely to be an external rather than an internal factor, because the variation between days follows the same pattern within a cohort. To control for these factors in a future experiment, all animals should be run in the same cohort and preferably in an isolated room to reduce the effects of the aforementioned factors.

The stabilisation of intake, preference and total fluid intake in the 24 h access period is probably because a measurement over a longer period is less subject to fluctuations in the external environment. Moreover, during the light-phase the animals were not disturbed by human influences, because no experiments were run. Stabilisation rather than an increase in the consumed amount of ethanol is suggesting that animals are not dependent on the drug or at least the consumed amount is sufficient to maintain their dependency. Hence, the motivation to consume the ethanol might be due to the reinforcing properties of small doses of ethanol, which have a stimulating effect. While high doses of ethanol may act as a non-stimulant in isolated rats and weaken responsiveness to communication (Wolffgramm, 1990). However, the males used in this study do increase their intake after the expansion of the period that ethanol is available, but the preference remains the same and a stabilisation in intake could be observed.

The intermittent ethanol access model has some limitations. Only 50-80% of the animals show escalated ethanol drinking. Additionally half of these animals do not reach a Blood ethanol concentration (BEC) of >80 mg%, which is the legal limit for driving a motor vehicle. Thus, BEC levels are relatively

low in animals subject to an intermittent ethanol access schedule.

Ethanol doses ranging from 2.5-60% induce an inverted U-shaped dose-response curve, with 20% being the optimum. The level of ethanol consumption seems to be tightly regulated to achieve a certain blood ethanol concentration. The preference for the ethanol solution might be a reflection of the BEC, because the preference remains the same throughout the experiment. Intake of 0.8-1.0 g/kg ethanol induced a blood ethanol level (BEC) of 60 mg% (Carnicella et al., 2011). This BEC is considered within the legal intoxicating range and often has a stimulating effect.

The ethanol-naïve ethanol-preferring P-rats showed reduced levels of dopamine and 5-HT in the brain compared to their nonpreferring counterparts (Bell, Rodd, Lumeng, Murphy, & McBride, 2006). This is in contrast to the innate increased synaptic levels of 5-HT found in our *SERT*^{-/-} animals. These neurochemical differences could explain the relatively low intake of ethanol in the *SERT*^{-/-} and *SERT*^{+/+} observed in this study. Moreover, pharmacological increases in 5-HT or dopamine decreased ethanol consumption in several different animal models (McBride & Li, 1998).

From approximately 30 days of age males become heavier compared to females (Pahl, 1969). Animals used for this study were adult and therefore at least 3 months of age explaining the significant difference in body weight. Moreover, heavier animals consume more food and fluids compared to smaller animals accounting for differences in total fluid intake. However, the relative fluid intake, calculated as the total fluid intake divided by the bodyweight, was higher for females compared to males. This may be due to the fact that the females are smaller and require a higher water intake (Krohn, Farlov, & Hansen, 2011). On the other hand, there was a positive correlation between fluid intake and bodyweight. However, the oestrous cycle might play a role for the fluid intake in females.

The animals that were used for voluntary ethanol consumption were individually housed, while rats are known to be social animals. The isolated environment is likely to induce stress and drug use of the animals, because a social environment reduced intake of morphine (Alexander, Coombs, & Hadaway, 1978) and ethanol (Wolffgramm, 1990). The free choice of ethanol and water used in this study can contribute to the lower intake of ethanol. However, to measure drinking behaviour per individual animal it was necessary to isolate animals, because our main goal was to investigate whether SERT^{-/-} would be differentially influenced by the reinforcing properties of ethanol. Moreover, the animals could still perform an intermediate degree of social interaction by communication through the bars of the lid of the cage (Wolffgramm, 1990), reducing stress and ethanol intake. The animals used by Simms et al. (2008) were individually housed in ventilated Plexiglas cages, implicating a higher degree of isolation and hence a higher degree of stress and drug use. Since ethanol has an anxiolytic effect and the escalated drinking may relieve stress due to single housing or other factors. For individual rats the cost of drinking is fairly low compared to humans. In humans, the work performance, family situation, and the socioeconomic status can be affected. The ethanol drinking in rats could even have a positive effect for the animals instead of the assumed negative effects of ethanol abuse (Hopf & Lesscher, 2014). There was no escalation of ethanol intake observed in this study and the animals did not seem to be extremely stressed. It is therefore likely that the results reflect the rat equivalent of recreational ethanol intake, rather than ethanol addiction or dependence.

Experiment 2 – Ethanol-induced locomotor activity

Locomotor activity was decreased by a medium dose of ethanol, but lower doses had no influence. This is in line with a previous

study reporting no differences in the effects of low doses of ethanol in rats. It was found that doses of 0.1 to 1.0 g/kg ethanol had no effect and that doses of 1.5-3.0 g/kg decreased locomotor activity in Sprague-Dawley rats (Frye & Breese, 1981). Other studies found that low doses of ethanol (0.25 and 0.5) stimulate locomotor behaviour (Agabio et al., 2001; Carlsson, Engel, & Svensson, 1972; Imperato & Di Chiara, 1986; Waller, Murphy, McBride, Lumeng, & Li, 1986) and increase dopamine release in the nucleus accumbens (Imperato & Di Chiara, 1986). However, the increase in locomotor activity was only seen in ethanol-preferring rats, but not in alcohol-nonpreferring animals (Agabio et al., 2001; Krimmer & Schechter, 1992; Quintanilla, 1999; Waller et al., 1986). Since our animals show a low preference for the ethanol solution during the voluntary drinking sessions they are comparable to the alcohol-nonpreferring animals. Moreover, the alcohol-nonpreferring animals show a higher baseline activity, which is also seen in the SERT^{-/-} animals in this study.

Ethanol-induced locomotor activity might be dependent on higher levels of stress. The animals were tested in a novel environment, which can be viewed as a stressful experience. Hypoactivity in a novel environment is seen as a sign of anxiety (Agabio et al., 2001). However, the SERT^{-/-} animals show hyperactivity in the open field, but do not show a difference in time spent in the centre. In other tests for anxiety they have been reported to show anxious behaviour (Olivier et al., 2008). The first half of the animals were injected intraperitoneal with 100% ethanol, because the animals were relatively heavy. The high bodyweight implicated that large volumes were necessary to reach the higher doses. A study suggested that it was better to keep the concentration constant and vary the volume of ethanol injected (Linakis & Cunningham, 1979). Concentration did not have an effect on total number of ambulatory counts in this study, which allowed us to combine the animals that

received 100% and 50% ethanol, because after more than 10% of the animals died following a dose of 2 g/kg this dose was excluded for further use and the concentration of ethanol was adjusted to 50%. In our study, survival was positively correlated with ethanol dose, and negatively with concentration. Thus a higher concentration and a higher dose were associated with a higher mortality. Intraperitoneal injection of 100% ethanol could induce mild to moderate inflammation of the peritoneum. This inflammation can last for up to two weeks after injection. Moreover, for humans a maximum of 1 mL 100% ethanol/kg is determined to be safe (In, Kim, Park, & Kim, 2014). The dose of 2 g/kg ethanol used in this study exceeds this guideline. Moreover, leakage of the ethanol to surrounding tissue could induce necrosis of this tissue (In et al., 2014). Besides, ethanol is a dehydrating agent and most animals that did not survive were severely dehydrated. A study in mice showed that a dose of 3.2 ml/kg 100% ethanol is lethal for the entire test population of 10 animals (Worthley & Schott, 1965). 3.2 ml/kg is equal to 2.5 g/kg. There is no obvious explanation for the fact that several animals that received the 2.0 g/kg died, most likely animals died as a consequence of one or several of the aforementioned complications. No correlation was found between survival and genotype, which suggests that there was no differential sensitivity for ethanol between the genotypes. However, the 2.0 g/kg dose is widely used in other literature without any reports on mortality (Carlsson et al., 1972; Frye & Breese, 1981; Imperato & Di Chiara, 1986). The difference with these studies is that they used a concentration of 20% ethanol. Therefore, it is most likely that the concentration of 100% ethanol is too high and that it would be better to inject bigger volumes of lower concentrations to reach a high dose.

Experiment 3 – Ethanol-induced ataxia

SERT^{-/-} animals are likely to be less sensitive for ethanol-induced ataxia compared to SERT^{+/+} and SERT^{+/-} animals. Moreover, SERT^{-/-} animals showed a better performance on the rotarod under baseline conditions. The higher activity of the SERT^{-/-} might contribute to the better performance of these animals. The results are contrary to findings in the SERT^{-/-} mouse model, which showed a shorter latency to fall from the accelerating rotarod (Boyce-Rustay et al., 2006).

Training of the animals to use the rotarod requires a certain degree of skill and patience. Once the animals had an adverse experience, they refused to go onto the rod in subsequent trials. Moreover, it is tricky to keep them motivated to stay on it and not jump or slide off. Individual differences were notable during the training sessions, but a criterion of a 210 s minimum on the rotarod did not influence the data substantially.

A concentration of up to 2.0 g/kg should be sufficient to result in ethanol-induced ataxia. Since a BEC of 122 mg% is able to induce ataxia in rodents (Arrasate, Mitra, Schweitzer, Segal, & Finkbeiner, 2004). This is the case for the SERT^{+/+} and SERT^{+/-} animals, but not for the SERT^{-/-} animals as concluded from the relative rotarod data. The higher baseline performance could indicate a bigger buffer to compensate for the ataxic effects of ethanol. Besides, a differential metabolic efficiency of ethanol could be an underlying cause for the observed difference at 2.0 g/kg. Unfortunately, we were not yet able to determine the BEC, due to technical issues (see appendix for discussion). Therefore, future research is necessary to elucidate this possibility.

General discussion

Research in macaque monkeys showed that chronic ethanol intake results in a decrease of SERT density in the hippocampus. This effect may be a response to the decreased 5-HT levels following chronic ethanol consumption (Burnett, Davenport, Grant, & Friedman, 2012). However, 5-HT has a great variety of

targets, with each possibly a differential effect on reward (Hayes & Greenshaw, 2011). The SERT^{-/-} rats showed increased baseline levels of 5-HT in the extracellular space of the brain (J. Homberg et al., 2007). This could lead to an increased activity of 5-HT neurotransmission and on the other hand, a desensitization of the receptors. The 5-HT transporter density is high in the raphe nuclei, ventral tegmental area, and somewhat lower in the striatum and the nucleus accumbens (Kranz et al., 2010).

The weak effect of genotype on ethanol intake in humans is subject to several confounding factors, like socioeconomic status and major life events, which makes a study using an animal model easier to interpret. Interestingly, heterozygous SERT knock-out rats (SERT^{+/-}) show, comparable to humans homozygous for the S-allele of the 5-HTT, a 50% reduction in 5-HT transporter ability (J. Homberg et al., 2007). This makes the SERT^{+/-} a valuable subject to include in the study, because it increases translational validity to the human situation.

Only 60% of the risk to become addicted can be explained by genetics and so far research has failed to find a specific gene that explains why approximately 10-20% of people who try ethanol actually get addicted. The variance explained by gender, level of education and 5-HTTLPR genotype together was only 16%, and the effect of the 5-HTTLPR should therefore not be over-interpreted (van der Zwaluw et al., 2010). Indeed, addiction is a complex heterogeneous phenomenon with a lot of different elements involved, amongst which are early life stress and metabolic enzymes like the aldehyde dehydrogenase 2

(ALDH2) enzyme, which is responsible for ethanol break down (Szalavitz, 2015). In humans, it is likely that the S-allele alters the effect of ethanol intake instead of causing alcoholism by itself (Andreas Heinz et al., 2000). Moreover, several different personality traits are associated with addiction. Examples are anxiety and impulsivity. The current study was unable to detect differences in anxious behaviour between the SERT^{-/-}, SERT^{+/-} and SERT^{+/+}, but earlier studies showed that the SERT^{-/-} spent less time in the centre of an open field and in the open arms of the elevated plus maze, and displayed a longer latency to start feeding and leave the home cage in a novel environment (Olivier et al., 2008). Impulsivity in SERT^{-/-} is investigated using the Five-choice serial reaction time task and did not differ from the performance of their heterozygous and wild-type counterparts (J. R. Homberg et al., 2007). As shown in this study, the different genotypes did not show a differential sensitivity for the anxiolytic properties of ethanol as well as a differential anxious baseline measure.

Environmental factors are as influential on the phenotype as genetic factors and can eventually even overrule genetics by having an effect on the epigenetics (Szalavitz, 2015). It might be that the genetic reduction in the serotonin transporter only leads to an increased sensitivity for ethanol addiction in combination with environmental factors, such as early life stress. The effect of early life stress on ethanol consumption would be an interesting topic for a future study.

References

- Agabio, R., Carai, M. A. M., Lobina, C., Pani, M., Reali, R., Vacca, G., . . . Colombo, G. (2001). Alcohol stimulates motor activity in selectively bred Sardinian alcohol-preferring (sP), but not in Sardinian alcohol-nonpreferring (sNP), rats. *Alcohol*, 23(2), 123-126.
- Alexander, B. K., Coombs, R. B., & Hadaway, P. F. (1978). The effect of housing and gender on morphine self-administration in rats. *Psychopharmacology*, 58(2), 175-179.
- Arrasate, M., Mitra, S., Schweitzer, E. S., Segal, M. R., & Finkbeiner, S. (2004). Inclusion body formation reduces levels of mutant huntingtin and the risk of neuronal death. *Nature*, 431(7010), 805-810.
- Babor, T. F., Higgins-Biddle, J. C., Saunders, J. B., & Monteiro, M. G. (2001). *Audit: The Alcohol Use Disorders Identification Test (AUDIT): Guidelines for use in primary care*. World Health Organization.
- Bell, R. L., Rodd, Z. A., Lumeng, L., Murphy, J. M., & McBride, W. J. (2006). REVIEW: The alcohol-preferring P rat and animal models of excessive alcohol drinking. *Addiction Biology*, 11(3-4), 270-288.
- Boyce-Rustay, J. M., Wiedholz, L. M., Millstein, R. A., Carroll, J., Murphy, D. L., Daws, L. C., & Holmes, A. (2006). Ethanol-related behaviors in serotonin transporter knockout mice. *Alcoholism: Clinical and Experimental Research*, 30(12), 1957-1965.
- Brodie, M. S., Trifunović, R. D., & Shefner, S. A. (1995). Serotonin potentiates ethanol-induced excitation of ventral tegmental area neurons in brain slices from three different rat strains. *Journal of Pharmacology and Experimental Therapeutics*, 273(3), 1139-1146.
- Burnett, E. J., Davenport, A. T., Grant, K. A., & Friedman, D. P. (2012). The effects of chronic ethanol self-administration on hippocampal serotonin transporter density in monkeys. *Frontiers in Psychiatry*, 3, 38.
- Caillé, S., Alvarez-Jaimes, L., Polis, I., Stouffer, D. G., & Parsons, L. H. (2007). Specific Alterations of Extracellular Endocannabinoid Levels in the Nucleus Accumbens by Ethanol, Heroin, and Cocaine Self-Administration. *The Journal of Neuroscience*, 27(14), 3695-3702.
- Cao, J., Hudziak, J. J., & Li, D. (2013). Multi-Cultural Association of the Serotonin Transporter Gene (SLC6A4) with Substance Use Disorder. *Neuropsychopharmacology*, 38(9), 1737-1747.
- Carlsson, A., Engel, J., & Svensson, T. H. (1972). Inhibition of ethanol-induced excitation in mice and rats by α -methyl-p-tyrosine. *Psychopharmacologia*, 26(3), 307-312.
- Carnicella, S., Ron, D., & Barak, S. (2014). Intermittent ethanol access schedule in rats as a preclinical model of alcohol abuse. *Alcohol*, 48(3), 243-252.
- Carnicella, S., Yowell, Q. V., & Ron, D. (2011). Regulation of Operant Oral Ethanol Self-Administration: A Dose-Response Curve Study in Rats. *Alcoholism: Clinical and Experimental Research*, 35(1), 116-125.
- Chemel, B. R., Roth, B. L., Armbruster, B., Watts, V. J., & Nichols, D. E. (2006). WAY-100635 is a potent dopamine D4 receptor agonist. *Psychopharmacology*, 188(2), 244-251.
- Daws, L. C., Montañez, S., Munn, J. L., Owens, W. A., Baganz, N. L., Boyce-Rustay, J. M., . . . Holmes, A. (2006). Ethanol inhibits clearance of brain serotonin by a serotonin transporter-independent mechanism. *The Journal of Neuroscience*, 26(24), 6431-6438.
- Diamond, I., & Gordon, A. S. (1997). Cellular and molecular neuroscience of alcoholism. *Physiological Reviews*, 77(1), 1-20.
- Fabre, V., Beaufour, C., Evrard, A., Rioux, A., Hanoun, N., Lesch, K., . . . Martres, M.

- P. (2000). Altered expression and functions of serotonin 5-HT_{1A} and 5-HT_{1B} receptors in knock-out mice lacking the 5-HT transporter. *European Journal of Neuroscience*, 12(7), 2299-2310.
- Feinn, R., Nellissery, M., & Kranzler, H. R. (2005). Meta-analysis of the association of a functional serotonin transporter promoter polymorphism with alcohol dependence. *American Journal of Medical Genetics Part B: Neuropsychiatric Genetics*, 133B(1), 79-84.
- Frye, G. D., & Breese, G. R. (1981). An evaluation of the locomotor stimulating action of ethanol in rats and mice. *Psychopharmacology*, 75(4), 372-379.
- Gobbi, G., Murphy, D. L., Lesch, K.-P., & Blier, P. (2001). Modifications of the serotonergic system in mice lacking serotonin transporters: an in vivo electrophysiological study. *Journal of Pharmacology and Experimental Therapeutics*, 296(3), 987-995.
- Hayes, D. J., & Greenshaw, A. J. (2011). 5-HT receptors and reward-related behaviour: a review. *Neuroscience & biobehavioral reviews*, 35(6), 1419-1449.
- Heils, A., Teufel, A., Petri, S., Stöber, G., Riederer, P., Bengel, D., & Lesch, K. P. (1996). Allelic variation of human serotonin transporter gene expression. *Journal of neurochemistry*, 66(6), 2621-2624.
- Heinz, A., Ragan, P., Jones, D. W., Hommer, D., Williams, W., Knable, M. B., . . . Linnoila, M. (1998). Reduced Central Serotonin Transporters in Alcoholism. *American Journal of Psychiatry*, 155(11), 1544-1549.
- Heinz, A., Jones, D. W., Gorey, J. G., Bennet, A., Suomi, S. J., Weinberger, D. R., & Higley, J. D. (2003). Serotonin transporter availability correlates with alcohol intake in non-human primates. *Molecular Psychiatry*, 8(2), 231-234.
- Heinz, A., Jones, D. W., Mazzanti, C., Goldman, D., Ragan, P., Hommer, D., . . . Weinberger, D. R. (2000). A relationship between serotonin transporter genotype and in vivo protein expression and alcohol neurotoxicity. *Biological psychiatry*, 47(7), 643-649.
- Hillmer, A. T., Wooten, D. W., Tudorascu, D. L., Barnhart, T. E., Ahlers, E. O., Resch, L. M., . . . Schneider, M. L. (2014). The effects of chronic alcohol self-administration on serotonin-1A receptor binding in nonhuman primates. *Drug and alcohol dependence*, 144, 119-126.
- Homberg, J., Olivier, J., Smits, B., Mul, J., Mudde, J., Verheul, M., . . . Cremers, T. (2007). Characterization of the serotonin transporter knockout rat: a selective change in the functioning of the serotonergic system. *Neuroscience*, 146(4), 1662-1676.
- Homberg, J. R., De Boer, S. F., Raasø, H. S., Olivier, J. D., Verheul, M., Ronken, E., . . . Vanderschuren, L. J. (2008). Adaptations in pre-and postsynaptic 5-HT_{1A} receptor function and cocaine supersensitivity in serotonin transporter knockout rats. *Psychopharmacology*, 200(3), 367-380.
- Homberg, J. R., Pattij, T., Janssen, M. C., Ronken, E., De Boer, S. F., Schoffemeer, A. N., & Cuppen, E. (2007). Serotonin transporter deficiency in rats improves inhibitory control but not behavioural flexibility. *European Journal of Neuroscience*, 26(7), 2066-2073.
- Hopf, F. W., & Lesscher, H. M. (2014). Rodent models for compulsive alcohol intake. *Alcohol*, 48(3), 253-264.
- Imperato, A., & Di Chiara, G. (1986). Preferential stimulation of dopamine release in the nucleus accumbens of freely moving rats by ethanol. *Journal of Pharmacology and Experimental Therapeutics*, 239(1), 219-228.
- In, H. S., Kim, D. W., Park, Y. M., & Kim, B. (2014). Experimental intraperitoneal

- injection of alcohol in rats: Peritoneal findings and histopathology. *Toxicology Reports*, 1, 31-35.
- Jacobs, E. H., Smit, A. B., de Vries, T. J., & Schoffelmeer, A. N. M. (2003). Neuroadaptive effects of active versus passive drug administration in addiction research. *Trends in Pharmacological Sciences*, 24(11), 566-573.
- Kirby, L., Zeeb, F., & Winstanley, C. (2011). Contributions of serotonin in addiction vulnerability. *Neuropharmacology*, 61(3), 421-432.
- Koob, G. F., & Le Moal, M. (2005). Plasticity of reward neurocircuitry and the 'dark side' of drug addiction. *Nature neuroscience*, 8(11), 1442-1444.
- Koob, G. F., Sanna, P. P., & Bloom, F. E. (1998). Neuroscience of addiction. *Neuron*, 21(3), 467-476.
- Koob, G. F., & Volkow, N. D. (2010). Neurocircuitry of Addiction. *Neuropsychopharmacology*, 35(1), 217-238.
- Kranz, G., Kasper, S., & Lanzenberger, R. (2010). Reward and the serotonergic system. *Neuroscience*, 166(4), 1023-1035.
- Krimmer, E. C., & Schechter, M. D. (1992). HAD and LAD rats respond differently to stimulating effect but not discriminative effects of ethanol. *Alcohol*, 9(1), 71-74.
- Krohn, T. C., Farlov, H., & Hansen, A. (2011). The effects of housing on food and water consumption in da-rats. *Scandinavian Journal of Laboratory Animal Sciences*, 38(1), 39-44.
- Kuhar, M. J., Ritz, M. C., & Boja, J. W. (1991). The dopamine hypothesis of the reinforcing properties of cocaine. *Trends in Neurosciences*, 14(7), 299-302.
- Lamb, R., & Daws, L. (2013). Ethanol self-administration in serotonin transporter knockout mice: unconstrained demand and elasticity. *Genes, Brain and Behavior*, 12(7), 741-747.
- Lesch, K.-P., Bengel, D., Heils, A., Sabol, S. Z., Greenberg, B. D., Petri, S., . . . Murphy, D. L. (1996). Association of anxiety-related traits with a polymorphism in the serotonin transporter gene regulatory region. *Science*, 274(5292), 1527-1531.
- Linakis, J. G., & Cunningham, C. L. (1979). Effects of concentration of ethanol injected intraperitoneally on taste aversion, body temperature, and activity. *Psychopharmacology*, 64(1), 61-65.
- Little, K. Y., McLaughlin, D. P., Zhang, L., Livermore, C. S., Dalack, G. W., McFinton, P. R., . . . Watson, S. J. (1998). Cocaine, ethanol, and genotype effects on human midbrain serotonin transporter binding sites and mRNA levels. *American Journal of Psychiatry*, 155(2), 207-213.
- Lovinger, D. M. (1997). Serotonin's role in alcohol's effects on the brain. *Alcohol health and research world*, 21, 114-119.
- McBride, W. J., & Li, T.-K. (1998). Animal models of alcoholism: neurobiology of high alcohol-drinking behavior in rodents. *Critical Reviews™ in Neurobiology*, 12(4), 339-369.
- Meisch, R., & Thompson, T. (1971). Ethanol intake in the absence of concurrent food reinforcement. *Psychopharmacologia*, 22(1), 72-79.
- Meredith, R. (2015). Sensitive and critical periods during neurotypical and aberrant neurodevelopment: A framework for neurodevelopmental disorders. *Neuroscience & biobehavioral reviews*, 50, 180-188.
- Moonat, S., Starkman, B. G., Sakharkar, A., & Pandey, S. C. (2010). Neuroscience of alcoholism: molecular and cellular mechanisms. *Cellular and Molecular Life Sciences*, 67(1), 73-88.
- Oakly, A., Brox, B., Schenk, S., & Ellenbroek, B. (2014). A genetic deletion of the serotonin transporter greatly enhances the reinforcing properties of MDMA in rats. *Molecular psychiatry*, 19(5), 534-535.

- Olivier, J., Van Der Hart, M., Van Swelm, R., Dederen, P., Homberg, J., Cremers, T., . . . Ellenbroek, B. (2008). A study in male and female 5-HT transporter knockout rats: an animal model for anxiety and depression disorders. *Neuroscience*, 152(3), 573-584.
- Pahl, P. (1969). Growth curves for body weight of the laboratory rat. *Australian journal of biological sciences*, 22(4), 1077-1080.
- Prus, A. J., James, J. R., & Rosecrans, J. A. (2009). *Chapter 4. Conditioned Place Preference*. (2nd edition. ed.). Boca Raton (FL): CRC Press.
- Quintanilla, M. E. (1999). Effect of low doses of ethanol on spontaneous locomotor activity in UChB and UChA rats. *Addiction Biology*, 4(4), 443-448.
- Simms, J. A., Steensland, P., Medina, B., Abernathy, K. E., Chandler, L. J., Wise, R., & Bartlett, S. E. (2008). Intermittent Access to 20% Ethanol Induces High Ethanol Consumption in Long-Evans and Wistar Rats. *Alcoholism: Clinical and Experimental Research*, 32(10), 1816-1823.
- Smits, B. M., Mudde, J. B., van de Belt, J., Verheul, M., Olivier, J., Homberg, J., . . . Plasterk, R. H. (2006). Generation of gene knockouts and mutant models in the laboratory rat by ENU-driven target-selected mutagenesis. *Pharmacogenetics and genomics*, 16(3), 159-169.
- Szalavitz, M. (2015). Genetics: No more addictive personality. *Nature*, 522(7557), S48-S49.
- van der Zwaluw, C. S., Engels, R. C., Vermulst, A. A., Rose, R. J., Verkes, R. J., Buitelaar, J., . . . Scholte, R. H. (2010). A serotonin transporter polymorphism (5-HTTLPR) predicts the development of adolescent alcohol use. *Drug and alcohol dependence*, 112(1), 134-139.
- Vengeliene, V., Bilbao, A., Molander, A., & Spanagel, R. (2008). Neuropharmacology of alcohol addiction. *British journal of pharmacology*, 154(2), 299-315.
- Waller, M. B., Murphy, J. M., McBride, W. J., Lumeng, L., & Li, T. K. (1986). Effect of low dose ethanol on spontaneous motor activity in alcohol-preferring and -nonpreferring lines of rats. *Pharmacology Biochemistry and Behavior*, 24(3), 617-623.
- Weiss, F., Parsons, L. H., Schulteis, G., Hyytiä, P., Lorang, M. T., Bloom, F. E., & Koob, G. F. (1996). Ethanol self-administration restores withdrawal-associated deficiencies in accumbal dopamine and 5-hydroxytryptamine release in dependent rats. *The Journal of Neuroscience*, 16(10), 3474-3485.
- Wise, R. A. (1973). Voluntary ethanol intake in rats following exposure to ethanol on various schedules. *Psychopharmacologia*, 29(3), 203-210.
- Wolffgramm, J. (1990). Free choice ethanol intake of laboratory rats under different social conditions. *Psychopharmacology*, 101(2), 233-239.
- Worthley, E. G., & Schott, C. D. (1965). Pharmacotoxic evaluation of nine vehicles administered intraperitoneally to mice. Maryland: US Army Edgewood arsenal.

Appendix

Blood Ethanol Concentration

Material & methods

After the last measurement of motor coordination on the rotating roll, animals were euthanized using CO₂ inhalation in order to perform an intracardiac blood collection. The animals were placed on their back after holding it vertical by its front legs for a moment to allow the heart to fall in a natural position. A 22 G needle was inserted just to the left of the base of the sternum and a vacuum was created by pulling back the syringe plunger. After reaching the heart, approximately 0.9 ml blood was collected. The collected blood was transferred to 500 µl K₂EDTA coated BD microtainers (365974, Becton, Dickinson and Company, Franklin Lakes, USA) and centrifuged for 10 min on 1300 g. The resulting supernatant was carefully aspirated and stored at -80°C until analysis. The ethanol assay kit (MAK076, Sigma-Aldrich, St. Louis, USA) was used to measure blood ethanol concentration. The kit contains ethanol assay buffer, ethanol probe, in DMSO, ethanol enzyme mix and an ethanol standard of 17.15 N. Reactions were performed in 96 well flat-bottom plates and incubated for 60 minutes at room temperature protected from light. Fluorometric ($\lambda_{\text{ex}} = 535/\lambda_{\text{em}} = 587 \text{ nm}$) and colorimetric detection of the absorbance at 570 nm (A_{570}) were performed by a multiwall plate reader (2300 Enspire Multimode Plate Reader, Perkin Elmer, Waltham, Massachusetts, USA). Concentration of ethanol could be calculated from the results of the assay by the formula $S_a/S_v = C$, where S_a is the amount of ethanol in the sample, S_v is sample volume and C is concentration of ethanol in sample.

Results

Ethanol concentration in the blood after i.p injection was meant to be measured by colorimetric analysis, but we were only able to generate a standard curve with the formula

$y = 0.0572x - 0.0275$ and $R^2 = 0.9924$. The pilot sample dilutions did not yield reliable results (not shown).

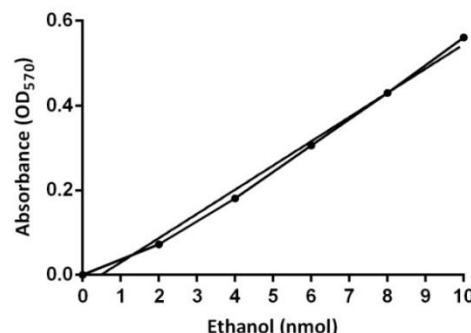


Figure 3. Standard curve ethanol assay. Standard curve generated with the diluted ethanol standard. Values are expressed with background signal subtracted. $y = 0.0572x - 0.0275$ and $R^2 = 0.9924$.

Discussion

The ethanol assay kit used to measure blood ethanol concentration is extremely sensitive for the presence of short-chain alcohols in the environment. The kit uses the conversion of ethanol by alcohol oxidase to generate hydrogen peroxide (H₂O₂) which reacts with the probe to generate colour. Moreover, the probe used is highly sensitive to oxygen and moisture, resulting in oxidation of the probe before addition to the ethanol-containing samples. There were some problems with the measurement. First of all, the fluorescence assay indicated a ceiling effect in the output of the standard curve. The next two standard curves set up for the colorimetric assay looked good, hence the decision to use the colorimetric assay for the samples. However, the next couple of trials the standard curve did not work anymore, which means that the outcomes for the samples are unfit for use.

After exclusion of several possible explanations the explanations that still hold are; decrease of the probe integrity following repeated freeze/thaw cycles, which can be prevented by preparing aliquots of the probe after the first thawing. Moreover, the integrity of the enzyme mix could be reduced by inappropriate storage resulting in no protection from light.

Protocol ethanol self-administration

Materials

- Graduated cylinder
- 1 L bottle
- 99.8% pure ethanol
- Tap water
- 120 mL graduated drinking bottles (Living World #61530)
- Scale

Procedure

- Prepare 20% ethanol solution by diluting 200 mL 99.8% ethanol with 800 mL tap water. Each animal should have at least 50 mL 20% ethanol per session. (Solution can be made in advance and stored in a sealable bottle. New solution should be prepared for each drinking session, because evaporation can take place).
 - Tare scale for an empty bottle.
 - 20 min after lights-off, bottles that are on the cage are weighted on the scale and bottles that are not needed are put aside to be washed.
 - Pour 20% ethanol in each bottle, weight bottle and present to animal.
 - After 7 h access weight bottles and animals, take away ethanol if animals are in 7 h access phase and replace with water bottle. Put ethanol bottle back if animals are in 24 h phase.
 - After 24 h access weight bottles again and set aside the bottles that are not needed. Replace ethanol bottles with water bottles.
 - Repeat this cycle Monday, Wednesday and Friday for 20% ethanol and Tuesday, Thursday and Saturday for water, alternating the placement side of the ethanol bottle in consecutive sessions.
- (- If possible, have an empty cage with an ethanol and water bottle to measure spillage between each weighting moment).

Protocol locomotion behaviour

- Bring animals into the testing room at least **10 minutes** prior to testing to allow them to adjust to the environment.
- Set up the Activity monitor program with the right settings, Resolution (ms): 50, Box Size: 4, Resting Delay (ms): 500, Session Time (min): 30, Chamber Type: ENV-510, Units: Cm.
- Inject rats with saline, 0.25, 0.5 or 1.0 g/kg of 50% ethanol adjusted for body weight.
- Place rats in open field for the duration of 30 min.
- After all rats are finished, get them out of the boxes and clean it with dissolving Virkon S in water (1:100).
- Analyse the data by defining the zones and exporting it to a text file. This can be transferred to a portable disk and used for analysis.

Protocol rotarod

- Bring animals into the testing room at least **10 minutes prior** to training/testing to allow them to adjust to the environment.

Serotonin transporter and effects of ethanol

- Make sure the RotaRod unit is plugged in (it is usually unplugged after use), the on/off switch is located on the back left-hand side of the unit.
- The front screen should now be illuminated, "TIME" should read 0:00 for each of the four slots. "SPEED" should read 0 rpm, and "MODE" should read Stop.
- Prior to beginning your experiment the switch on the front, left-hand side of the unit should be in the "STOP" position (bottom-most position).
- With the switch you can produce a fixed speed rotation (using "RUN", the topmost position). Speed (rpm) is controlled by the knob on the right-hand side of the unit "SPEED (RUN)".
- You can produce an accelerating rotation (using "ACC", the left-hand switch in the middle position). The cylinder will accelerate from 4 – 40 rpm, the knob on the left controls the time over which the acceleration will take place.

Training:

- Set RotaRod to fixed speed ("RUN"; the goal is to build up to 12 rpm, see point "i" below).
- Give each rat 3 trials a day for 3 days (maximum of 3 min. with a 5 minute intertrial interval).
- i. Gently place animal onto rotating cylinder, and raise the white lever below the cylinder to start trial. When you raise the white lever the time for that specific slot will start. It will continue to count up until the rat falls off the cylinder and stops the timer. Record this time (and the rpm) in your logbook. On Day 1 start the cylinder at a low speed (4 rpm) and increase the speed gradually.
- Record latency to fall and speed for each trial.
- Clean apparatus with 70% ethanol (make sure to be very careful cleaning around the switch and under the lever, urine and feces can build up here and cause the switch to break).

Testing:

- **48 h** after the last training trial inject rats i.p. with 0, 1.0 or 2.0 g/kg of ethanol. **10 and 30 min** after ethanol administration record motor behaviour. Set RotaRod to accelerating mode ("ACC"; 4 – 40 rpm over 5 minutes) and raise the white levers. The cylinder will start rotating at 4 rpm.
- Place rat on rod and push "start" button. The timers will start, and the rpm will be displayed for each slot. If you haven't raised the levers, you will be prompted to do so before the start of the experiment (WARNING: Lift the needed levers).
- Give each rat two trials (maximum time 5 minutes). If a subject remained on the rod until this time they were removed and a score of 360 s was assigned.
- Record latency to fall, and rpm at which animal fell for each trial.
- **Following the 30 min trial**, collect blood for blood ethanol concentration (BEC) assay.
- When finished, clean apparatus and testing area with 70% ethanol. Sweep the floor if necessary.

Protocol blood plasma preparation

Materials and Equipment

- Blood sample
- Vacutainer tubes containing anticoagulant (e.g. BD Vacutainer plastic EDTA tube, 500 µl, lavender top)
- Pipettes of appropriate volumes (sterile)
- Centrifuge tubes
- Benchtop centrifuge

Procedure

- Draw blood into vacutainer tube(s) containing ~1.8 mg K2EDTA per ml blood (may vary depending on manufacturer) (lavender tops). Be sure to draw the full volume to ensure the correct blood-to-anticoagulant ratio.
- Invert vacutainer tubes carefully 10 times to mix blood and anticoagulant and store at room temperature until centrifugation.
- Samples should undergo centrifugation immediately or at least within 30 min. This should be carried out for a minimum of 10 minutes at 1300 RCF (or x g) at room temperature. Do not use brake to stop centrifuge.
- This will give three layers: (from top to bottom) plasma, leucocytes (buffy coat), and erythrocytes.
- Carefully aspirate the supernatant (plasma) at room temperature and pool in an Eppendorf tube and mix by pipetting. Take care not to disrupt the cell layer or transfer any cells.
- Inspect plasma for turbidity. Turbid samples should be centrifuged and aspirated again to remove remaining insoluble matter.
- Aliquot 100 µL plasma into Eppendorf tubes and store at –80 °C. Ensure that the tubes are adequately labelled with the relevant information.

Protocol Ethanol Essay Kit

Components

The kit is sufficient for 100 assays in 96 well plates.

Ethanol Assay Buffer	25 mL
----------------------	-------

Catalog Number MAK076A

Ethanol Probe, in DMSO	0.2 mL
------------------------	--------

Catalog Number MAK076B

Ethanol Enzyme Mix	1 vL
--------------------	------

Catalog Number MAK076C

Ethanol Standard, 17.15 N	0.5 mL
---------------------------	--------

Catalog Number MAK076D

Serotonin transporter and effects of ethanol

Equipment Required

- 96 well flat-bottom plate – It is recommended to use black plates with clear bottoms for fluorescence assays and clear plates for colorimetric assays.
- Fluorescence multiwell plate reader

Preparation Instructions

- Briefly **centrifuge** vials before opening.
- To maintain reagent integrity, avoid repeated freeze/thaw cycles
- Ethanol Assay Buffer – Allow buffer to come to **room temperature** before use.
- Ethanol Probe – Warm to room temperature to melt frozen solution prior to use. **Aliquot** and store **protected from light and moisture at –20 °C**.
 - o For the fluorescence assay, dilute an aliquot of the Ethanol Probe Solution **5 to 10-fold** with Ethanol Assay Buffer, just prior to use. This will reduce the background of the fluorescence assay.
- Ethanol Enzyme Mix – Reconstitute in 220 mL of Ethanol Assay Buffer. Mix well by pipetting, then store, **protected from light**, at 2–8 °C. Stable for 2 months.

Procedure

This assay should **not** be performed in areas where alcohol-containing solvents are stored or where laboratory benches are wiped down with alcohol.

All samples and standards should be run in **duplicate**.

Ethanol Standards for Fluorometric Detection

- Generate a 10 nmole/ μ L ethanol standard solution:
 - o Dilute 50 μ L of the 17.15 N Ethanol Standard with 808.7 μ L of the Ethanol Assay Buffer to generate a 1 μ mole/ μ L standard.
 - o Dilute 10 μ L of the 1 μ mole/ μ L standard solution with 990 μ L of ethanol assay buffer to generate a 10 nmole/ μ L solution.
- Dilute 10 μ L of the 10 nmole/ μ L ethanol standard with 990 μ L of the Ethanol Assay Buffer to generate a 0.1 nmole/ μ L standard solution.
- Add 0, 2, 4, 6, 8, and 10 μ L of the 0.1 nmole/ μ L standard solution into a 96 well plate, generating 0 (blank), 0.2, 0.4, 0.6, 0.8, and 1.0 nmole/well standards.
- Add Ethanol Assay Buffer to each well to bring the volume to **50 μ L**.

Ethanol standards for colorimetric detection

- Generate a 10 nmole/ μ L ethanol standard solution:
 - o Dilute 50 μ L of the 17.15 N Ethanol Standard with 808.7 μ L of the Ethanol Assay Buffer to generate a 1 μ mole/ μ L standard.
 - o Dilute 10 μ L of the 1 μ mole/ μ L standard solution with 990 μ L of ethanol assay buffer to generate a 10 nmole/ μ L solution.
- Dilute 100 μ L of the 10 nmole/ μ L standard solution with 900 μ L of ethanol assay buffer to generate a 1 nmole/ μ L solution.

Serotonin transporter and effects of ethanol

- Add 0, 2, 4, 6, 8, and 10 μL of the 1 nmole/ μL standard solution into a 96 well plate, generating 0 (blank), 2, 4, 6, 8, and 10 nmole/well standards.
- Add Ethanol Assay Buffer to each well to bring the volume to **50 μL** .

Sample Preparation

- Samples should be diluted in Ethanol Assay Buffer.
- Bring samples to a final volume of **50 μL** with Ethanol Assay Buffer.
- For 1.0 g/kg samples dilute 10 μL of sample with 90 μL Ethanol Assay Buffer (10x dilution). Dilute 10 μL of this dilution in 90 μL Ethanol Assay Buffer (100x).
- For 2.0 g/kg samples dilute 10 μL of sample with 90 μL Ethanol Assay Buffer (10x). Dilute 10 μL of this dilution with 90 μL Ethanol Assay Buffer (100x dilution).

Assay Reaction

1. Set up the Master Reaction Mix according to the scheme in Table 1. 50 μL of the Master Reaction Mix is required for each reaction (well).

Table 1. Reaction Mix

Reagent	Volume per well	Volume per plate (42 samples)	Volume per half plate (18 samples)	Volume for 36 samples
Ethanol Assay Buffer	46 μL	4416 μL = 4.416 mL	2208 μL = 2.208 mL	3864 μL
Ethanol Probe	2 μL	192 μL	96 μL	168 μL
Ethanol Enzyme Mix	2 μL	192 μL	96 μL	168 μL
Total	50 μL	4800 μL = 4.8 mL	2400 μL 2.4 mL	4200 μL = 4.2 mL

2. Add **50 μL** of the Master Reaction Mix to each of the wells. Mix well using a horizontal shaker or by pipetting, and incubate the reaction for 30 minutes at 37 °C or 60 minutes at room temperature. **Cover** the plate tightly and protect the plate from light during the incubation.
3. For colorimetric assays, measure the absorbance at 570 nm (A_{570}).
For fluorometric assays, measure fluorescence intensity ($\lambda_{\text{ex}} = 535/\lambda_{\text{em}} = 587 \text{ nm}$).

Results

Calculations

- The background for the assays is the value obtained for the 0 (blank) Ethanol Standard.
- Correct for the background by subtracting the 0 (blank) value from all readings. Background values can be significant and must be subtracted from all readings.
- Use the values obtained from the appropriate ethanol standards to plot a standard curve.

Note: A new standard curve must be set up each time the assay is run.

Subtract the blank value from the sample reading to obtain the corrected measurement.

Using the corrected measurement, the amount of ethanol present in the sample may be determined from the standard curve.

Serotonin transporter and effects of ethanol

Concentration of Ethanol

$$Sa/Sv = C$$

Sa = Amount of ethanol in unknown sample (nmole) from standard curve

Sv = Sample volume (μL) added into the wells

C = Concentration of ethanol in sample

Ethanol molecular weight: **46.07 g/mole**

Sample Calculation

Amount of ethanol (Sa) = 5.84 nmole (from standard curve)

Sample volume (Sv) = 50 μL

Concentration of ethanol in sample

$$5.84 \text{ nmole}/50 \text{ μL} = 0.1168 \text{ nmole}/\text{μL}$$

$$0.1168 \text{ nmole}/\text{μL} * 46.07 \text{ ng/nmole} = 5.38 \text{ ng}/\text{μL}$$

$$5.38 \text{ ng}/\text{μL} = 0.538 \text{ mg/dl}$$